

Part A. Exploring and Preparing the Data

- Observations and Variables: There are 53940 observations and 10 variables.
- Response Variable: The commodity price is the response variable, and it measures the diamond price in US dollars.
- Quantitative Explanatory Variable: The carat is the quantitative explanatory variable, and it measures the weight of the diamond.
- Categorical Explanatory Variable: The cut variable is the categorical explanatory variable, and it represents the diamond's cutting quality. It has five categories, namely, Fair, Good, Very Good, Premium, and Ideal.
- Reference Category: "Ideal" has been selected as the reference category, which is necessary for dummy variable usage to avoid the problem of perfect multicollinearity.

Part B. Constructing Dummy Variables

- **Dummy-Variable Coding Table** Using Ideal as the reference category, the coding scheme is:

Cut	D1 (Fair)	D2 (Good)	D3 (Very Good)	D4 (Premium)
Fair	1	0	0	0
Good	0	1	0	0
Very Good	0	0	1	0
Premium	0	0	0	1
Ideal	0	0	0	0

- The All-Zero Category: The all-zero category represents the "Ideal" cut baseline. When a diamond is Ideal, \$D_1\$ through \$D_4\$ are all 0, meaning the model's intercept represents the baseline value for an Ideal diamond.
- The Intercept and Five Dummies: Five dummy variables should not be included together with an intercept because their sum would always equal 1, creating a perfect linear dependency with the intercept. This makes the mathematical matrix non-invertible, preventing the regression software from estimating unique coefficients.

Part C. Additive Dummy-Variable Regression

Estimated Regression Equation:

$$\hat{Y}_i = [\text{Intercept}] + [\text{carat}]X_i + [\text{cutFair}]D_{1i} + [\text{cutGood}]D_{2i} + [\text{cutVery Good}]D_{3i} + [\text{cutPremium}]D_{4i}.$$

Interpret Carat: Holding the cut quality constant, for every 1-unit increase in carat weight, the estimated price of the diamond increases by [carat coef] US dollars, on average.

Interpret Dummy (e.g., Fair): Holding carat weight constant, a diamond with a Fair cut is estimated to cost [cutFair coef] dollars less/more than a diamond with an Ideal cut, on average.

Interpret R^2 : The R^2 value of [Multiple R-squared] indicates that approximately $X\%$ of the total variance in diamond prices is explained by the combination of carat weight and cut quality in this additive model.

Slopes: The regression lines for the different cut categories have the same slope. An additive dummy-variable model assumes the effect of the quantitative predictor (carat) is constant across all categories, shifting only the intercept (starting price) for each cut.

Part D. Interaction Between Carat and Cut

Hypotheses:

- H_0 : There is no interaction between carat and cut; the interaction coefficients are equal to zero ($H_0: \gamma_{interaction} = 0$).
- H_A : There is an interaction between carat and cut; at least one interaction coefficient is not equal to zero.

Interaction Coefficients: Look at the summary(model2) output for variables with a colon (e.g., carat:cutFair). Note their estimates and their corresponding p-values (which will likely be < 0.05).

ANOVA/Incremental F-test: The anova(model1, model2) output compares the additive and interaction models. Note the F-statistic and the p-value provided in the output table.

Statistical Decision: Because the p-value from the ANOVA test is < 0.05 , reject the null hypothesis. In the context of diamond prices, this means the relationship between carat weight and price is significantly different depending on the cut quality.

Practical Difference: The additive model assumes that an additional carat adds the exact same dollar amount to a diamond's price regardless of its cut. The interaction model acknowledges that an additional carat increases the price at *different rates* depending on the cut (e.g., a larger carat increases price much faster for an Ideal cut than for a Fair cut).

Part E. Visualization and Statistical Conclusion

The regression analysis demonstrates a strong positive relationship between carat weight and price; as diamonds get larger, they become substantially more expensive. The role of cut serves as a significant modifier to this baseline relationship. By utilizing dummy variables, we established the Ideal cut as our comparative baseline, revealing the price penalties associated with lower-quality cuts. Crucially, the interaction model and the accompanying scatterplot prove that the effect of carat differs significantly among cut categories. The non-parallel regression lines in the visual plot indicate that the market places a steeper price premium on carat weight when the diamond possesses a high-quality cut. Ultimately, diamond pricing is highly dependent not just on the individual effects of weight and cut, but on how those two features multiply together.